25900707

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACAQC200 - AQUATIC POLLUTION AND MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Name any two major sources of aquatic pollution. [K] / [CO1]
2. Name any two pollutants that belong to the category of agricultural waste. [K] / [CO2]
3. Write two examples for biodegradable waste. [K] / [CO4]
4. Name one source of nitrogen in aquatic ecosystems. [U] / [CO3]
5. Write a source that contribute iron in aquatic ecosystem. [U] / [CO3]
6. What is the objective of The Environment (Protection) Act? [K] / [CO4]
7. What happens to submerged vegetation during algal bloom? [U] / [CO2]
8. Which index categorizes organisms by their tolerance to organic pollution? [K] / [CO4]
9. Define microbial pollution. [U] / [CO2]
10. Give one example of a toxic organic compound found in polluted water. [U] / [CO1]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11.	What are the major impacts of thermal pollution in aquatic environment?	[U] / [CO2]
12.	Explain how industrial effluents pollute aquatic ecosystems.	[U] / [CO3]
13.	Differentiate between natural and man-made aquatic pollution.	[K] / [CO1]
14.	Why is bioremediation considered a natural process?	[U] / [CO5]
15.	Define Chemical Oxygen Demand (COD) and explain why it is important.	[U] / [CO3]
16.	Comment on the impact of toxic organic compounds on aquatic organisms	[K] / [CO1]
17.	Why are red tides more frequent in warm coastal waters?	[U] / [CO3]
18.	Differentiate between bioaugmentation and biostimulation	[U] / [CO5]
19.	Explain how heavy metals enter aquatic ecosystem.	[U] / [CO2]
20.	Write any two limitations of water quality index	[K] / [CO4]
21.	Write any two factors which affect the efficiency of bioremediation	[U] / [CO5]
22.	How does pesticide accumulation in fish pose risks to human consumers?	[K] / [CO2]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

23.	Explain any five disadvantages of phytoremediation	[U] / [CO5]
24.	Explain how sewage reduces the dissolved oxygen content of water.	[U] / [CO2]
25.	Describe oil pollution – sources and its impact on aquatic ecosystems	[K] / [CO1]
26.	Explain how chemical pollutants affect the quality of water and aquatic life.	[U] / [CO1]
27.	What are the control and preventive measures of oil pollution?	[U] / [CO2]
28.	Discuss the biological productivity differences between eutrophic and oligotrophic lakes.	[U] / [CO2]

[5x4 = 20]